Enhanced Index Strategy based on S&P 500 Sub-Index

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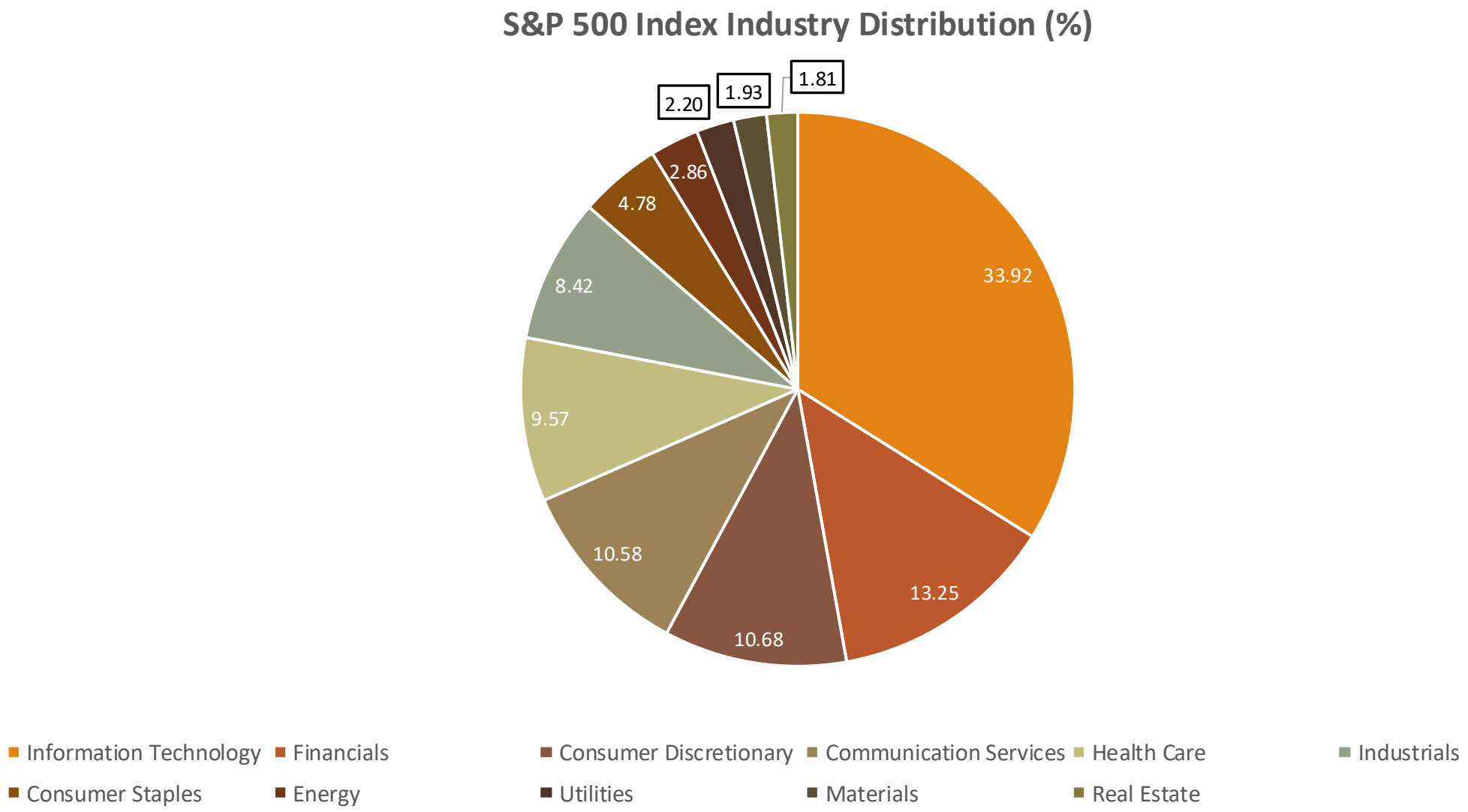
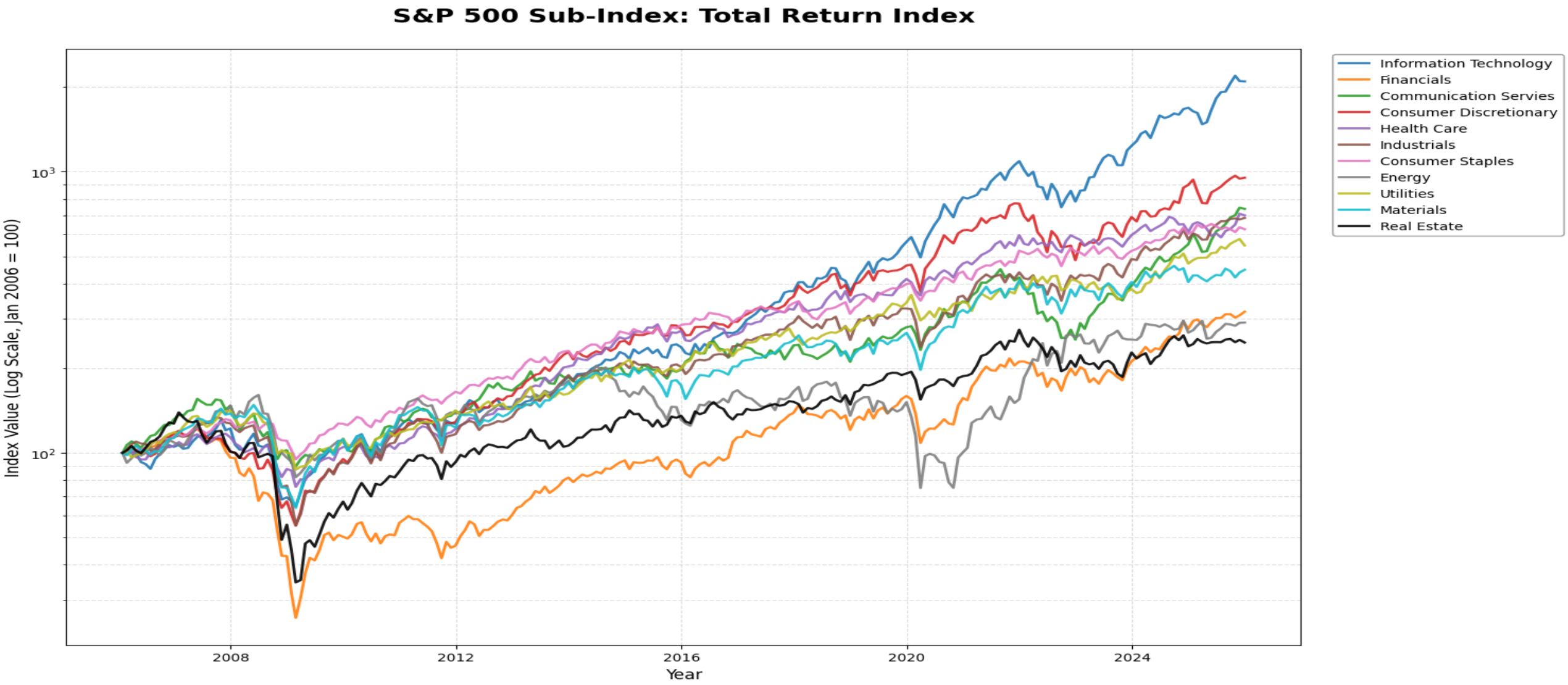
S&P 500 Industry Distribution



S&P 500 Sub-Index



Macro Regime Framework

Lagged Correlation	Information Technology	Financials	Communication Services	Consumer Discretionary	Health Care	Industrials	Consumer Staples	Energy	Utilities	Materials	Real Estate
ΔFFR	9.68%	12.77%	8.60%	7.97%	10.90%	14.40%	10.13%	20.91%	13.48%	12.10%	10.25%
ΔFFR (t-1)	0.02%	13.29%	-4.09%	-0.92%	-3.54%	8.05%	5.25%	-8.08%	0.29%	-2.88%	11.22%
ΔCPI (t-1)	-6.33%	-4.41%	-10.42%	-7.50%	1.40%	-5.24%	1.47%	-5.81%	1.49%	-7.47%	-4.78%
ΔCPI (t-2)	-7.12%	7.97%	-6.15%	-1.64%	3.35%	2.55%	7.99%	-0.64%	3.30%	-2.80%	9.46%
ΔDXY (t-1)	-9.49%	1.83%	2.83%	0.21%	4.31%	-1.08%	3.14%	7.33%	-3.68%	3.52%	-5.19%

CPI Strategy

- Strategy 1: Choose 4 industries (Financials, Health Care, Consumer Staples, and Real Estate) that may have better return when ΔCPI (t-2) > 0 to the portfolio
- Strategy 2: Choose 4 industries (Information Technology, Communication Services, Consumer Discretionary, and Materials) that may have better return when ΔCPI (t-2) < 0 to the portfolio
- Dynamic Strategy: Shift to Strategy 1 if ΔCPI (t-2) > 0, while shift to Strategy 2 if ΔCPI (t-2) < 0

Statistic Table	Strategy 1	Strategy 2	Dynamic Strategy	SPX
Annualized Return	8.26%	12.29%	12.64%	10.99%
Annualized Volatility	15.07%	16.62%	15.56%	15.06%
Return/Vol	0.55	0.74	0.81	0.73
Maximum Drawdown	-57.64%	-49.44%	-39.42%	-50.91%
Strategy-Switching	x	x	86	x

FFR Strategy

- Strategy 1: Choose 3 industries (Financials, Industrials, and Real Estate) that may have better return when ΔFFR (t-1) > 0 to the portfolio
- Strategy 2: Choose 3 industries (Communication Services, Health Care, and Energy) that may have better return when ΔFFR (t-1) < 0 to the portfolio
- Dynamic Strategy: Shift to Strategy 1 if ΔFFR (t-1) > 0, while shift to Strategy 2 if ΔFFR (t-1) < 0

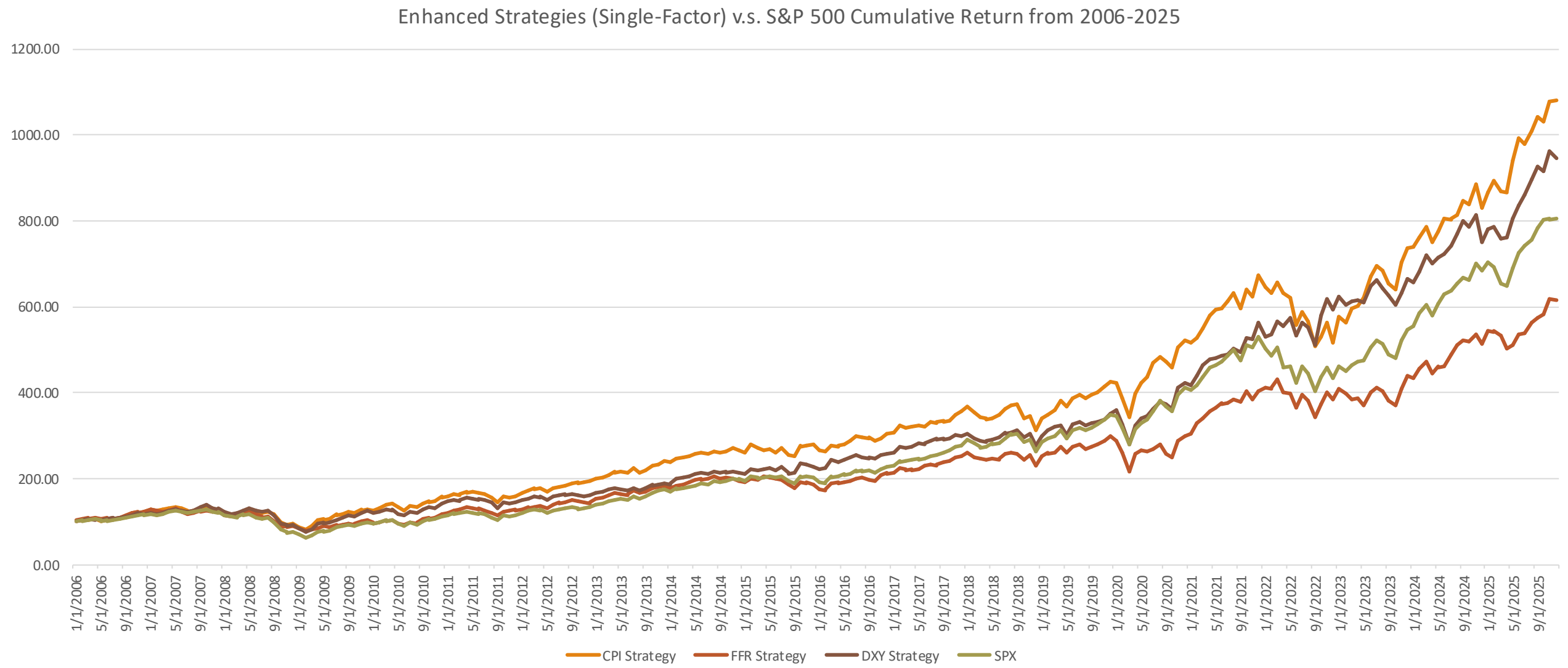
Statistic Table	Strategy 1	Strategy 2	Dynamic Strategy	SPX
Annualized Return	7.41%	9.89%	9.50%	10.99%
Annualized Volatility	19.24%	15.10%	15.45%	15.06%
Return/Vol	0.39	0.65	0.61	0.73
Maximum Drawdown	-70.27%	-38.60%	-40.35%	-50.91%
Strategy-Switching	x	x	5	x

DXY Strategy

- Strategy 1: Choose 4 industries (Energy, Health Care, Consumer Staples, and Materials) that may have better return when ΔDXY (t-1) > 0 to the portfolio
- Strategy 2: Choose 4 industries (Information Technology, Utilities, Real Estate, and Industrials) that may have better return when ΔDXY (t-1) < 0 to the portfolio
- Dynamic Strategy: Shift to Strategy 1 if ΔDXY (t-1) > 0, while shift to Strategy 2 if ΔDXY (t-1) < 0

Statistic Table	Strategy 1	Strategy 2	Dynamic Strategy	SPX
Annualized Return	9.20%	10.74%	11.90%	10.99%
Annualized Volatility	14.83%	15.50%	15.23%	15.06%
Return/Vol	0.62	0.69	0.78	0.73
Maximum Drawdown	-40.55%	-54.98%	-45.87%	-50.91%
Strategy-Switching	x	x	120	x

Cumulative Return



CPI + FFR Strategy

- Strategy Formula: $\text{Score}_{[i,t]} = \text{Corr}(\Delta\text{FFR}(t-1))_{[i,T]} * Z(\Delta\text{FFR}(t-1)) + \text{Corr}(\Delta\text{CPI}(t-2))_{[i,T]} * Z(\Delta\text{CPI}(t-2))$
- $\text{Corr}(\Delta\text{FFR}(t-1))_{[i,T]}$ & $\text{Corr}(\Delta\text{CPI}(t-2))_{[i,T]}$: 20-year Correlation Coefficient

→ Represent the 20-year historical correlation between industry *i* monthly return and macro indicator shifts, which quantifies the sensitivity of each sector's performance to policy and inflation cycles

- $Z(\Delta\text{FFR}(t-1))$ & $Z(\Delta\text{CPI}(t-2))$: Standardized Macro Indicator Changes over 240 months

→ Calculate the Z-score of the macro factor's change, which identifies the severity and direction of the current economic regime relative to history

- By multiplying correlation by standardized macro factors, the model aims to identify which industry's character best aligns with the current macro climate, predicting risk-adjusted outperformance

CPI + FFR Strategy

- For example, the score for 2006/01/31 Information Technology Industry will be 0.02% * 1.1677 + (-7.12%) * (-1.6892) = 0.1204, and the score for 2006/02/28 will be 0.02% * 1.1677+ (-7.12%) * (-0.2010) = 0.0145
- The score transition from Jan 2006 to Feb 2006 reflects that the buy signal for Information Technology Industry diminish due to the stabilization of disinflationary momentum

Correlation Coefficient Matrix

Lagged Correlation	Information Technology	Financials	Communication Services	Consumer Discretionary	Health Care
ΔFFR	9.68%	12.77%	8.60%	7.97%	10.90%
ΔFFR (t-1)	0.02%	13.29%	-4.09%	-0.92%	-3.54%
ΔCPI (t-1)	-6.33%	-4.41%	-10.42%	-7.50%	1.40%
ΔCPI (t-2)	-7.12%	7.97%	-6.15%	-1.64%	3.35%

Standardized Macro Factors

Factor Standardization	ΔFFR	Standardized ΔFFR	ΔCPI	Standardized ΔCPI
2005/11/29	0.25	1.1677	-0.8	-1.6892
2005/12/30	0.25	1.1677	-0.1	-0.2010
2006/1/31	0.25	1.1677	0.6	1.2871
2006/2/28	0	0.0000	-0.4	-0.8388
2006/3/31	0.25	1.1677	-0.2	-0.4136

Selection & Weighting Methods

- After calculating the score of each industry over a 240-month period, we will add 3 highest score's industry to our strategy's portfolio each month, concentrating on top-tier signals to filter out sector-specific noise
- We utilize different weighting methods to evaluate the portfolio's monthly return:

→ **Equal Weighting Monthly Return** = Top 1~3 industry score's monthly return / 3

→ **Score-Based Weighting Monthly Return** = Top 1 industry score's monthly return * (Top 1 industry score / Total of Top1~3 industry score) + Top 2 industry score's monthly return * (Top 2 industry score / Total of Top1~3 industry score) + Top 3 industry score's monthly return * (Top 3 industry score / Total of Top1~3 industry score)

Selection & Weighting Methods

- **Inverse-Volatility Weighting:**

→ Weight Return Calculation = Return $_i$ $[(1 / \sigma_i) / \Sigma(1 / \sigma_{top1\sim3})]$

→ Timing Baseline: $\boldsymbol{\sigma}$ is derived from industry monthly return's 12-month rolling look-back period (t-12 to t-1)

→ Meaning: This weighting is based on the realized risk of the past year, aiming to prevent high-volatility industries from dominating the portfolio's drawdown during market stress

- **Hybrid Weighting:**

→ Weight Return Calculation = Return $_i$ $[(\text{Score}_i / \sigma_i) / \Sigma\,(\text{Score}_{top1\sim3} / \sigma_{top1\sim3})]$

→ Meaning: This weighting prioritizes industries that offer the highest "Macro Reward" (Score) per unit of "Risk" (Volatility), creating the most effective return

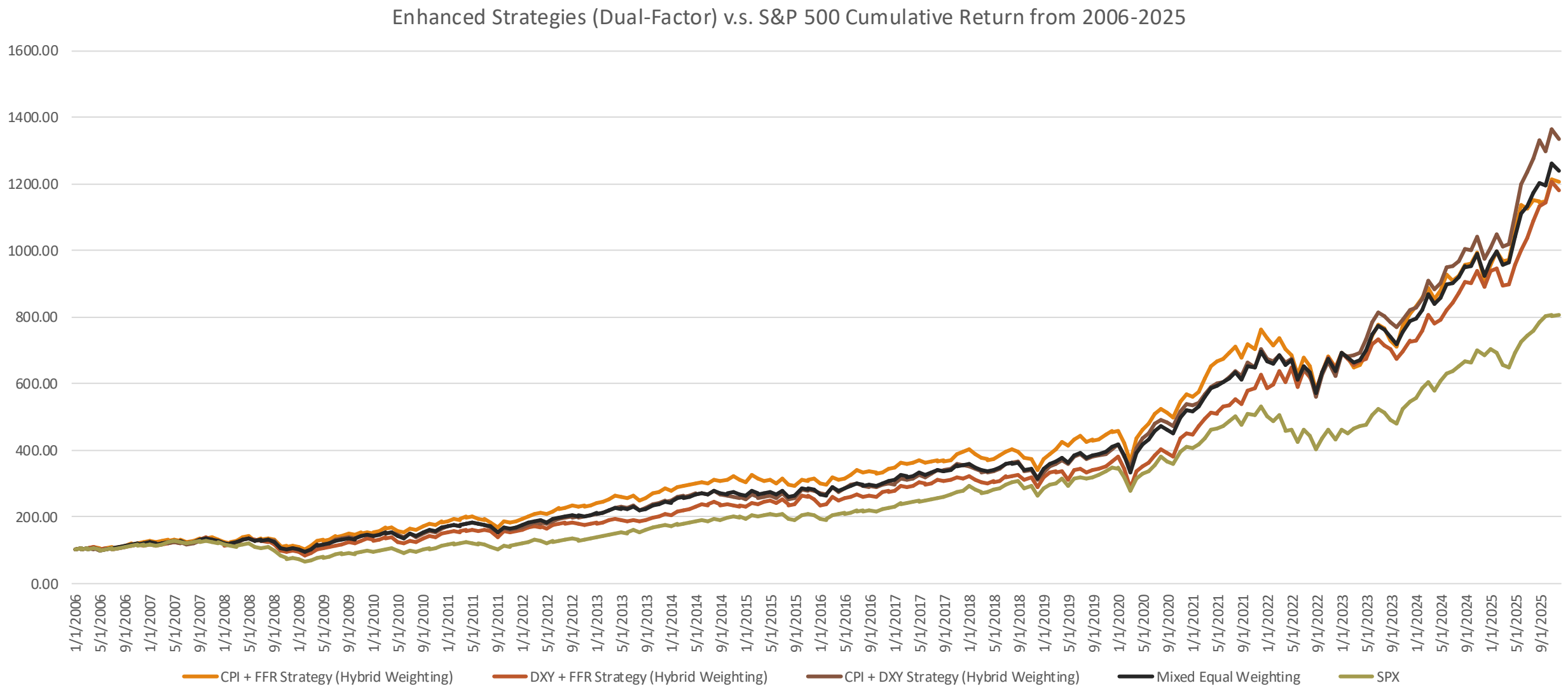
Statistic Table

	CPI + FFR Strategy (Equal Weighting)	CPI + FFR Strategy (Score-Based Weighting)	CPI + FFR Strategy (Inverse-Volatility Weighting)	CPI + FFR Strategy (Hybrid Weighting)	SPX
Annualized Return	12.81%	13.20%	12.69%	13.26%	10.99%
Annualized Volatility	15.92%	15.63%	15.03%	15.22%	15.06%
Return/Vol	0.80	0.85	0.84	0.87	0.73
Maximum Drawdown	-30.63%	-28.21%	-29.21%	-27.50%	-50.91%

Statistic Table

	Mixed Equal Weighting	CPI + FFR Strategy (Hybrid Weighting)	DXY + FFR Strategy (Hybrid Weighting)	CPI + DXY Strategy (Hybrid Weighting)	SPX
Annualized Return	13.42%	13.26%	13.14%	13.83%	10.99%
Annualized Volatility	14.95%	15.22%	16.22%	15.06%	15.06%
Return/Vol	0.90	0.87	0.81	0.92	0.73
Maximum Drawdown	-31.75%	-27.50%	-39.25%	-30.73%	-50.91%

Cumulative Return

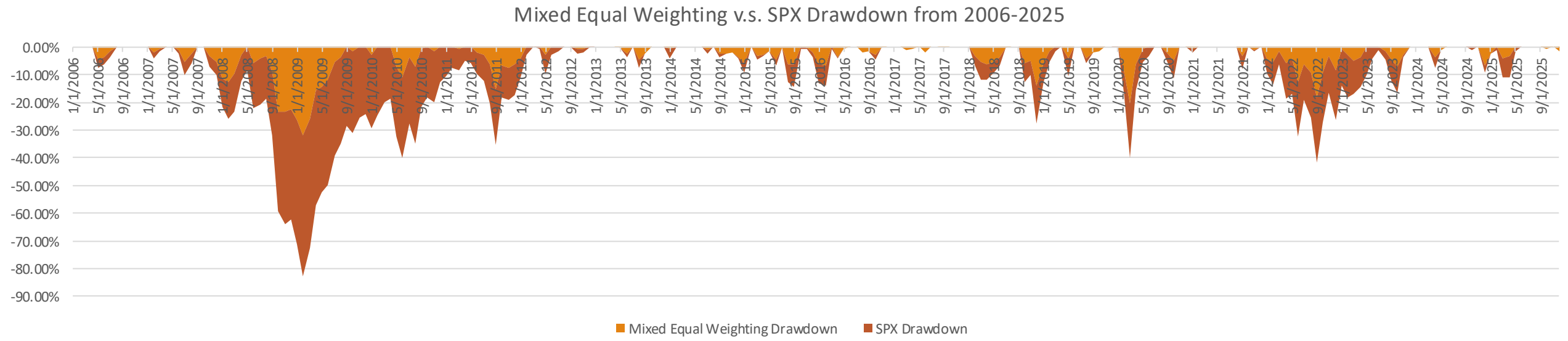


Hybrid Weighting Strategy

- Strength:

→ Higher Efficiency: It achieves a higher annualized return of 13.42%, outperforming the SPX's 10.99% over a 20-year horizon

→ Drastic Downside Protection: Its Maximum Drawdown has a much better value of -31.75%, cutting the SPX's severe -50.91% loss nearly in half over the past 20 years



The Financial Crisis 2008-2009 (CPI + FFR)

- Market Context: Severe equity contraction with sharp drop in CPI and FFR cuts to near-zero levels
- Strategic Pivot:

→ As CPI plummeted during the second-half of 2008, the portfolio automatically rotated to industries having a negative correlation with ΔCPI

- Key Allocations:

→ Disinflation Hedges: Heavy weighting in Information Technology and Communication Services (total >80%)

→ Risk Exclusion: Minimal exposure to Real Estate and Financials during the peak of the crisis

Selected Industry & Weight (CPI + FFR)

	Sector 1	Weight	Sector 2	Weight	Sector 3	Weight
2008.01	Health Care	35.06%	Energy	25.20%	Consumer Staples	39.73%
2008.02	Energy	45.14%	Communication Services	27.04%	Health Care	27.82%
2008.03	Real Estate	29.27%	Consumer Staples	44.80%	Financials	25.93%
2008.04	Energy	45.18%	Communication Services	27.27%	Materials	27.55%
2008.05	Energy	42.28%	Communication Services	21.47%	Health Care	36.24%
2008.06	Information Technology	39.19%	Communication Services	37.45%	Materials	23.36%
2008.07	Real Estate	29.86%	Consumer Staples	46.60%	Financials	23.55%
2008.08	Real Estate	31.28%	Consumer Staples	46.87%	Financials	21.85%
2008.09	Real Estate	31.98%	Consumer Staples	46.16%	Financials	21.87%
2008.10	Information Technology	42.21%	Communication Services	38.32%	Materials	19.47%
2008.11	Energy	42.11%	Communication Services	37.40%	Materials	20.49%
2008.12	Information Technology	43.10%	Communication Services	40.52%	Materials	16.38%
2009.01	Communication Services	43.88%	Information Technology	32.16%	Energy	23.95%
2009.02	Information Technology	44.17%	Communication Services	39.78%	Materials	16.04%
2009.03	Information Technology	43.20%	Communication Services	40.63%	Materials	16.17%
2009.04	Real Estate	22.97%	Consumer Staples	56.88%	Financials	20.14%
2009.05	Information Technology	40.22%	Communication Services	46.45%	Materials	13.33%
2009.06	Information Technology	40.16%	Communication Services	46.87%	Materials	12.97%

Selected Industry & Weight (CPI + FFR)

	Sector 1	Return	Sector 2	Return	Sector 3	Return	∑Return	SPX Return	ΔCPI (t-2)	ΔFFR (t-1)
2008.01	Health Care	-4.81	Energy	-10.92	Consumer Staples	-5.20	-6.51	-6.00	0.8	-0.25
2008.02	Energy	7.00	Communication Services	-9.60	Health Care	-2.26	-0.06	-3.25	-0.2	-1.25
2008.03	Real Estate	7.42	Consumer Staples	3.35	Financials	-2.73	2.97	-0.43	0.2	0
2008.04	Energy	10.87	Communication Services	4.81	Materials	5.46	7.72	4.87	-0.3	-0.75
2008.05	Energy	3.45	Communication Services	3.38	Health Care	2.02	2.92	1.30	0	-0.25
2008.06	Information Technology	-9.24	Communication Services	-11.52	Materials	-5.54	-9.23	-8.43	-0.1	0
2008.07	Real Estate	1.50	Consumer Staples	3.43	Financials	7.15	3.73	-0.84	0.3	0
2008.08	Real Estate	1.40	Consumer Staples	2.65	Financials	-1.08	1.44	1.45	0.8	0
2008.09	Real Estate	-2.01	Consumer Staples	-1.28	Financials	-4.89	-2.30	-8.91	0.6	0
2008.10	Information Technology	-17.79	Communication Services	-8.29	Materials	-22.07	-14.98	-16.80	-0.2	0
2008.11	Energy	0.77	Communication Services	6.49	Materials	-10.84	0.53	-7.17	-0.5	-1
2008.12	Information Technology	1.76	Communication Services	0.98	Materials	-0.37	1.10	1.06	-1.2	0
2009.01	Communication Services	-9.91	Information Technology	-3.05	Energy	-3.15	-6.08	-8.43	-2.6	-0.75
2009.02	Information Technology	-4.06	Communication Services	-2.80	Materials	-8.46	-4.26	-10.65	-1	0
2009.03	Information Technology	12.14	Communication Services	6.06	Materials	15.26	10.17	8.76	-0.1	0
2009.04	Real Estate	34.77	Consumer Staples	-0.74	Financials	22.36	14.36	9.57	0.2	0
2009.05	Information Technology	2.64	Communication Services	-1.28	Materials	5.82	1.24	5.59	-0.6	0
2009.06	Information Technology	4.06	Communication Services	1.14	Materials	-4.62	1.56	0.20	-0.3	0

The Financial Crisis 2008-2009 (CPI + DXY)

- Market Context: Severe equity contraction with sharp drop in CPI and a relatively high DXY rate
- Strategic Pivot:

→ As CPI plummeted and DXY rate rose during the second-half of 2008, the portfolio automatically rotated to industries having a negative correlation with ΔCPI and positive correlation with ΔDXY

- Key Allocations:

→ Disinflation Hedges: Heavy weighting in Information Technology and Communication Services

→ Strengthened USD: Incorporated industries like Energy and Materials to the portfolio

→ Risk Exclusion: Minimal exposure to Real Estate and Financials during the peak of the crisis

Selected Industry & Weight (CPI + DXY)

	Sector 1	Weight	Sector 2	Weight	Sector 3	Weight
2008.01	Real Estate	19.28%	Consumer Staples	52.27%	Financials	28.45%
2008.02	Information Technology	77.81%	Utilities	14.21%	Consumer Discretionary	7.98%
2008.03	Real Estate	48.35%	Utilities	29.67%	Information Technology	21.97%
2008.04	Information Technology	81.47%	Utilities	11.76%	Communication Services	6.77%
2008.05	Energy	34.71%	Health Care	39.92%	Materials	25.37%
2008.06	Communication Services	37.31%	Energy	27.80%	Materials	34.89%
2008.07	Real Estate	36.14%	Financials	22.30%	Consumer Staples	41.56%
2008.08	Consumer Staples	50.34%	Financials	22.67%	Real Estate	26.99%
2008.09	Consumer Staples	55.81%	Energy	22.06%	Financials	22.13%
2008.10	Energy	38.37%	Communication Services	31.21%	Materials	30.42%
2008.11	Energy	39.35%	Communication Services	34.13%	Materials	26.53%
2008.12	Communication Services	45.25%	Information Technology	34.10%	Materials	20.65%
2009.01	Information Technology	62.71%	Communication Services	28.64%	Consumer Discretionary	8.65%
2009.02	Communication Services	40.58%	Energy	33.44%	Materials	25.98%
2009.03	Energy	45.09%	Communication Services	29.34%	Materials	25.57%
2009.04	Real Estate	26.98%	Information Technology	37.65%	Utilities	35.38%
2009.05	Information Technology	54.85%	Communication Services	37.63%	Materials	7.52%
2009.06	Information Technology	65.02%	Real Estate	8.49%	Utilities	26.49%

Selected Industry & Weight (CPI + DXY)

	Sector 1	Return	Sector 2	Return	Sector 3	Return	∑Return	SPX Return	ΔCPI (t-2)	ΔDXY (t-1)
2008.01	Real Estate	-0.81	Consumer Staples	-5.20	Financials	-0.34	-2.97	-6.00	0.8	0.55
2008.02	Information Technology	-4.00	Utilities	-4.90	Consumer Discretionary	-3.91	-4.12	-3.25	-0.2	-1.52
2008.03	Real Estate	7.42	Utilities	1.67	Information Technology	0.95	4.29	-0.43	0.2	-1.46
2008.04	Information Technology	6.92	Utilities	5.28	Communication Services	4.81	6.59	4.87	-0.3	-1.91
2008.05	Energy	3.45	Health Care	2.02	Materials	4.80	3.22	1.30	0	0.71
2008.06	Communication Services	-11.52	Energy	2.28	Materials	-5.54	-5.60	-8.43	-0.1	0.37
2008.07	Real Estate	1.50	Financials	7.15	Consumer Staples	3.43	3.56	-0.84	0.3	-0.42
2008.08	Consumer Staples	2.65	Financials	-1.08	Real Estate	1.40	1.47	1.45	0.8	0.76
2008.09	Consumer Staples	-1.28	Energy	-11.96	Financials	-4.89	-4.44	-8.91	0.6	4.15
2008.10	Energy	-17.94	Communication Services	-8.29	Materials	-22.07	-16.19	-16.80	-0.2	2.07
2008.11	Energy	0.77	Communication Services	6.49	Materials	-10.84	-0.36	-7.17	-0.5	6.19
2008.12	Communication Services	0.98	Information Technology	1.76	Materials	-0.37	0.97	1.06	-1.2	0.88
2009.01	Information Technology	-3.05	Communication Services	-9.91	Consumer Discretionary	-10.56	-5.66	-8.43	-2.6	-5.21
2009.02	Communication Services	-2.80	Energy	-12.03	Materials	-8.46	-7.35	-10.65	-1	4.69
2009.03	Energy	3.79	Communication Services	6.06	Materials	15.26	7.39	8.76	-0.1	2.01
2009.04	Real Estate	34.77	Information Technology	12.08	Utilities	0.63	14.15	9.57	0.2	-2.58
2009.05	Information Technology	2.64	Communication Services	-1.28	Materials	5.82	1.40	5.59	-0.6	-0.82
2009.06	Information Technology	4.06	Real Estate	-5.21	Utilities	5.55	3.67	0.20	-0.3	-5.27

The Post-Pandemic Inflation 2022 (CPI + FFR)

- Market Context: Stagflationary environment where the market faced surging CPI and the most aggressive FFR hiking cycle in decades
- Strategic Pivot:

→ The model identified industries with a positive correlation to ΔCPI and ΔFFR to serve as portfolio anchors

- Key Allocations:

→ Inflation Hedges: High conviction in Financials (about 40%) and Real Estate (about 30%) during this period

→ Defensive Growth: Strategic use of Consumer Staples and Industrials to provide stability while growth sectors like Information Technology were devalued by rising macro indicators

Selected Industry & Weight (CPI + FFR)

	Sector 1	Weight	Sector 2	Weight	Sector 3	Weight
2022.01	Real Estate	38.39%	Consumer Staples	32.74%	Financials	28.87%
2022.02	Real Estate	32.05%	Consumer Staples	37.27%	Financials	30.68%
2022.03	Real Estate	29.08%	Consumer Staples	36.19%	Financials	34.73%
2022.04	Financials	43.59%	Real Estate	27.62%	Consumer Staples	28.79%
2022.05	Real Estate	28.97%	Consumer Staples	40.71%	Financials	30.32%
2022.06	Financials	44.04%	Real Estate	26.39%	Industrials	29.57%
2022.07	Financials	41.56%	Real Estate	30.97%	Industrials	27.47%
2022.08	Financials	42.06%	Real Estate	33.04%	Industrials	24.90%
2022.09	Information Technology	38.85%	Communication Services	43.92%	Materials	17.23%
2022.10	Financials	44.22%	Real Estate	28.50%	Industrials	27.28%
2022.11	Information Technology	38.43%	Communication Services	44.34%	Materials	17.23%
2022.12	Financials	43.17%	Real Estate	30.06%	Industrials	26.78%

Selected Industry & Weight (CPI + FFR)

	Sector 1	Return	Sector 2	Return	Sector 3	Return	∑Return	SPX Return	ΔCPI (t-2)	ΔFFR (t-1)
2022.01	Real Estate	-8.50	Consumer Staples	-1.37	Financials	0.06	-3.70	-5.17	0.6	0
2022.02	Real Estate	-4.91	Consumer Staples	-1.42	Financials	-1.35	-2.52	-2.99	0.2	0
2022.03	Real Estate	7.79	Consumer Staples	1.81	Financials	-0.19	2.86	3.71	0.5	0
2022.04	Financials	-9.87	Real Estate	-3.56	Consumer Staples	2.56	-4.55	-8.72	0.4	0.25
2022.05	Real Estate	-5.02	Consumer Staples	-4.61	Financials	2.73	-2.50	0.18	0.6	0
2022.06	Financials	-10.90	Real Estate	-6.90	Industrials	-7.40	-8.81	-8.26	-0.2	0.5
2022.07	Financials	7.21	Real Estate	8.54	Industrials	9.50	8.25	9.22	0.3	0.75
2022.08	Financials	-2.01	Real Estate	-5.61	Industrials	-2.81	-3.40	-4.08	0.5	0.75
2022.09	Information Technology	-12.01	Communication Services	-12.15	Materials	-9.35	-11.61	-9.21	-0.6	0
2022.10	Financials	11.99	Real Estate	2.04	Industrials	13.92	9.68	8.10	-0.2	0.75
2022.11	Information Technology	6.03	Communication Services	6.85	Materials	11.76	7.38	5.59	-0.1	0
2022.12	Financials	-5.27	Real Estate	-4.83	Industrials	-3.00	-4.53	-5.77	-0.5	0.75

The Post-Pandemic Inflation 2022 (CPI + DXY)

- Market Context: Stagflationary environment where the market faced surging CPI and DXY rate
- Strategic Pivot:

→ The model identified industries with a positive correlation to ΔCPI and ΔDXY to serve as portfolio anchors

- Key Allocations:

→ Inflation Hedges: High conviction in Financials (about 30%) during this period

→ Strengthened USD: Heavy Weighting in Consumer Staples and Health Care to avoid high volatility and provide steady return

Selected Industry & Weight (CPI + DXY)

	Sector 1	Weight	Sector 2	Weight	Sector 3	Weight
2022.01	Real Estate	41.01%	Financials	27.96%	Consumer Staples	31.02%
2022.02	Consumer Staples	42.92%	Financials	31.09%	Health Care	25.98%
2022.03	Real Estate	27.67%	Consumer Staples	37.13%	Financials	35.19%
2022.04	Consumer Staples	43.67%	Financials	33.90%	Health Care	22.42%
2022.05	Consumer Staples	48.57%	Energy	21.55%	Financials	29.88%
2022.06	Information Technology	82.85%	Communication Services	8.18%	Utilities	8.97%
2022.07	Energy	19.79%	Consumer Staples	47.03%	Health Care	33.18%
2022.08	Consumer Staples	49.42%	Financials	31.12%	Real Estate	19.46%
2022.09	Communication Services	45.71%	Energy	25.88%	Materials	28.42%
2022.10	Energy	35.63%	Communication Services	34.32%	Materials	30.05%
2022.11	Information Technology	79.43%	Communication Services	11.24%	Utilities	9.33%
2022.12	Information Technology	71.73%	Utilities	18.06%	Real Estate	10.20%

Selected Industry & Weight (CPI + DXY)

	Sector 1	Return	Sector 2	Return	Sector 3	Return	∑Return	SPX Return	ΔCPI (t-2)	ΔDXY (t-1)
2022.01	Real Estate	-8.50	Financials	0.06	Consumer Staples	-1.37	-3.90	-5.17	0.6	-0.32
2022.02	Consumer Staples	-1.42	Financials	-1.35	Health Care	-1.02	-1.29	-2.99	0.2	0.87
2022.03	Real Estate	7.79	Consumer Staples	1.81	Financials	-0.19	2.76	3.71	0.5	0.17
2022.04	Consumer Staples	2.56	Financials	-9.87	Health Care	-4.71	-3.28	-8.72	0.4	1.61
2022.05	Consumer Staples	-4.61	Energy	15.80	Financials	2.73	1.98	0.18	0.6	4.65
2022.06	Information Technology	-9.32	Communication Services	-7.69	Utilities	-4.98	-8.80	-8.26	-0.2	-1.21
2022.07	Energy	9.72	Consumer Staples	3.30	Health Care	3.32	4.58	9.22	0.3	2.93
2022.08	Consumer Staples	-1.75	Financials	-2.01	Real Estate	-5.61	-2.59	-4.08	0.5	1.22
2022.09	Communication Services	-12.15	Energy	-9.33	Materials	-9.35	-10.62	-9.21	-0.6	2.80
2022.10	Energy	24.96	Communication Services	0.14	Materials	9.00	11.65	8.10	-0.2	3.42
2022.11	Information Technology	6.03	Communication Services	6.85	Utilities	7.02	6.21	5.59	-0.1	-0.59
2022.12	Information Technology	-8.37	Utilities	-0.53	Real Estate	-4.83	-6.60	-5.77	-0.5	-5.58

Recent Year 2025 (CPI + FFR)

- Market Context: Transition to a FFR cutting cycle amidst sticky reflationary momentum
- Strategic Pivot:

→ The portfolio focused more on inflation-resilient anchors as FFR cuts started from the late 2024

- Key Allocations:

→ Inflation Hedges: Incorporated industries like Real Estate and Financials to the portfolio

→ Defensive Sectors: Strategic use of industries such as Consumer Staples and Health Care to manage overall portfolio volatility

Selected Industry & Weight (CPI + FFR)

	Sector 1	Weight	Sector 2	Weight	Sector 3	Weight
2025.01	Energy	43.91%	Health Care	30.37%	Communication Services	25.72%
2025.02	Real Estate	29.61%	Consumer Staples	42.19%	Financials	28.20%
2025.03	Real Estate	30.29%	Consumer Staples	40.41%	Financials	29.30%
2025.04	Information Technology	42.18%	Communication Services	38.06%	Materials	19.76%
2025.05	Information Technology	43.14%	Communication Services	37.24%	Materials	19.61%
2025.06	Information Technology	43.51%	Communication Services	36.15%	Materials	20.33%
2025.07	Real Estate	34.83%	Consumer Staples	37.36%	Financials	27.80%
2025.08	Real Estate	37.41%	Consumer Staples	34.86%	Financials	27.73%
2025.09	Real Estate	37.75%	Consumer Staples	36.17%	Financials	26.08%
2025.10	Energy	45.85%	Health Care	39.58%	Materials	14.57%
2025.11	Energy	46.13%	Health Care	34.69%	Communication Services	19.18%
2025.12	Energy	50.25%	Health Care	30.28%	Communication Services	19.47%

Selected Industry & Weight (CPI + FFR)

	Sector 1	Return	Sector 2	Return	Sector 3	Return	∑Return	SPX Return	ΔCPI (t-2)	ΔFFR (t-1)
2025.01	Energy	2.07	Health Care	6.79	Communication Services	9.12	5.32	2.78	0.1	-0.25
2025.02	Real Estate	4.23	Consumer Staples	5.70	Financials	1.41	4.05	-1.3	0.2	0
2025.03	Real Estate	-2.41	Consumer Staples	-2.43	Financials	-4.2	-2.94	-5.63	0.1	0
2025.04	Information Technology	1.62	Communication Services	0.75	Materials	-2.17	0.54	-0.68	-0.2	0
2025.05	Information Technology	10.89	Communication Services	9.63	Materials	3.03	8.88	6.29	-0.4	0
2025.06	Information Technology	9.77	Communication Services	7.28	Materials	2.32	7.36	5.08	-0.1	0
2025.07	Real Estate	-0.09	Consumer Staples	-2.37	Financials	-0.05	-0.93	2.24	0.1	0
2025.08	Real Estate	2.17	Consumer Staples	1.59	Financials	3.12	2.23	2.03	0.3	0
2025.09	Real Estate	0.49	Consumer Staples	-1.56	Financials	0.14	-0.34	3.64	0	0
2025.10	Energy	-1.13	Health Care	3.58	Materials	-5.04	0.16	2.34	0.2	-0.25
2025.11	Energy	2.49	Health Care	9.31	Communication Services	6.35	5.60	0.25	0.1	-0.25
2025.12	Energy	0.20	Health Care	-1.36	Communication Services	-0.98	-0.50	0.06	X	0

Recent Year 2025 (CPI + DXY)

- Market Context: Transition to a USD depreciation cycle amidst sticky reflationary momentum
- Strategic Pivot:

→ The portfolio focused more on industries having a negative correlation with DXY rate

- Key Allocations:

→ Weakened USD: Heavy Weighting in Information Technology (about 40%~70%)

→ Defensive Sectors: Strategic use of industries such as Utilities, Consumer Staples and Health Care to manage overall portfolio volatility

Selected Industry & Weight (CPI + DXY)

	Sector 1	Weight	Sector 2	Weight	Sector 3	Weight
2025.01	Energy	34.95%	Health Care	30.92%	Consumer Staples	34.13%
2025.02	Real Estate	32.18%	Financials	27.56%	Consumer Staples	40.26%
2025.03	Real Estate	46.69%	Information Technology	27.93%	Utilities	25.38%
2025.04	Information Technology	64.69%	Utilities	18.36%	Real Estate	16.95%
2025.05	Information Technology	71.82%	Utilities	15.62%	Real Estate	12.56%
2025.06	Information Technology	59.61%	Communication Services	29.11%	Materials	11.28%
2025.07	Information Technology	37.11%	Real Estate	37.26%	Utilities	25.63%
2025.08	Energy	27.11%	Consumer Staples	41.20%	Health Care	31.69%
2025.09	Information Technology	42.34%	Real Estate	34.29%	Utilities	23.36%
2025.10	Real Estate	38.65%	Consumer Staples	35.41%	Financials	25.94%
2025.11	Energy	30.69%	Health Care	30.78%	Consumer Staples	38.53%
2025.12	Information Technology	37.44%	Real Estate	37.20%	Utilities	25.36%

Selected Industry & Weight (CPI + DXY)

	Sector 1	Return	Sector 2	Return	Sector 3	Return	∑Return	SPX Return	ΔCPI (t-2)	ΔDXY (t-1)
2025.01	Energy	2.07	Health Care	6.79	Consumer Staples	2.04	3.52	2.78	0.1	2.75
2025.02	Real Estate	4.23	Financials	1.41	Consumer Staples	5.70	4.04	-1.3	0.2	-0.12
2025.03	Real Estate	-2.41	Information Technology	-8.83	Utilities	0.26	-3.53	-5.63	0.1	-0.76
2025.04	Information Technology	1.62	Utilities	0.10	Real Estate	-1.20	0.87	-0.68	-0.2	-3.40
2025.05	Information Technology	10.89	Utilities	3.83	Real Estate	0.99	8.54	6.29	-0.4	-4.74
2025.06	Information Technology	9.77	Communication Services	7.28	Materials	2.32	8.21	5.08	-0.1	-0.14
2025.07	Information Technology	5.19	Real Estate	-0.09	Utilities	4.94	3.16	2.24	0.1	-2.45
2025.08	Energy	3.64	Consumer Staples	1.59	Health Care	5.38	3.35	2.03	0.3	3.09
2025.09	Information Technology	7.25	Real Estate	0.49	Utilities	4.15	4.21	3.64	0	-2.20
2025.10	Real Estate	-2.63	Consumer Staples	-2.32	Financials	-2.84	-2.57	2.34	0.2	0.00
2025.11	Energy	2.49	Health Care	9.31	Consumer Staples	4.03	5.18	0.25	0.1	2.03
2025.12	Information Technology	-0.25	Real Estate	-2.19	Utilities	-5.12	-2.21	0.06	X	-0.34

Sub-Period Statistic Table

SPX	2008.01-2009.06	2022.01-2022.12	2025.01-2025.12
Period Return	-35.01%	-18.12%	17.86%
Annualized Return	-24.97%	-18.12%	17.86%
Annualized Volatility	23.99%	22.03%	10.52%
Maximum Drawdown	-50.91%	-23.87%	-7.50%

CPI + FFR Hybrid Weighting	2008.01-2009.06	2022.01-2022.12	2025.01-2025.12
Period Return	0.18%	-14.84%	32.78%
Annualized Return	0.12%	-14.84%	32.78%
Annualized Volatility	23.37%	22.38%	12.33%
Maximum Drawdown	-27.50%	-24.26%	-3.55%

CPI + DXY Hybrid Weighting	2008.01-2009.06	2022.01-2022.12	2025.01-2025.12
Period Return	-3.90%	-11.57%	37.05%
Annualized Return	-2.62%	-11.57%	37.05%
Annualized Volatility	22.85%	21.60%	13.04%
Maximum Drawdown	-30.73%	-20.16%	-3.53%

Conclusion & Future Outlook

- Superior Risk-Adjusted Performance:

→ Dual-Factor Strategies deliver robust annualized returns and significantly mitigate maximum drawdowns to reach a better performance than the SPX index

→ Dynamic sector rotation effectively captures macro-regime shifts by aligning portfolio sensitivity with CPI, DXY and FFR cycles

- Enhance Predictive Reliability:

→ Transition to a rolling-window correlation framework to eliminate look-ahead bias and reflect the time-varying macro sensitivities

→ Integrate more macro factors or labor market data to refine overall strategy's decision-making

→ Incorporate transaction costs, slippage and rebalancing frequency into the backtest to further validate strategy's net performance